

Robotics Setter Technician Program

Robotic welding, from learning to improvement

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Universidad Politécnica de Aguascalientes | Mechatronics Engineering, 9th term

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Engineering Department | August 1, 2026 to January 1, 2027

TENNECO

THE TENNECO VALUE I IDENTIFY WITH

Win

We must earn the trust of our employees and customers.

For me, winning means caring for the ergonomics of our welders while protecting product quality. As an engineer, I do it by optimizing the process with robotic systems.

Program scope and objective

Robotics Setter Technician | August 1, 2026 to January 1, 2027

OBJECTIVE

Become a specialist in robotic welding and advanced technical support.

WHY IT MATTERS

When a welding robot stops, the line stops with it. A setter who can find the root cause and adjust the cell on the spot keeps parts moving and quality under control.

FINAL PROJECT

An implemented improvement

In quality, availability or productivity.

Mine: 4M, implemented, with 47% less cycle time.

An updated training process

How the next setters learn.

Mine: study material reviewed, schedule rebuilt.

The plan: four levels, each one validated

My ILUO matrix, paraphrased | 40 hours of formal training

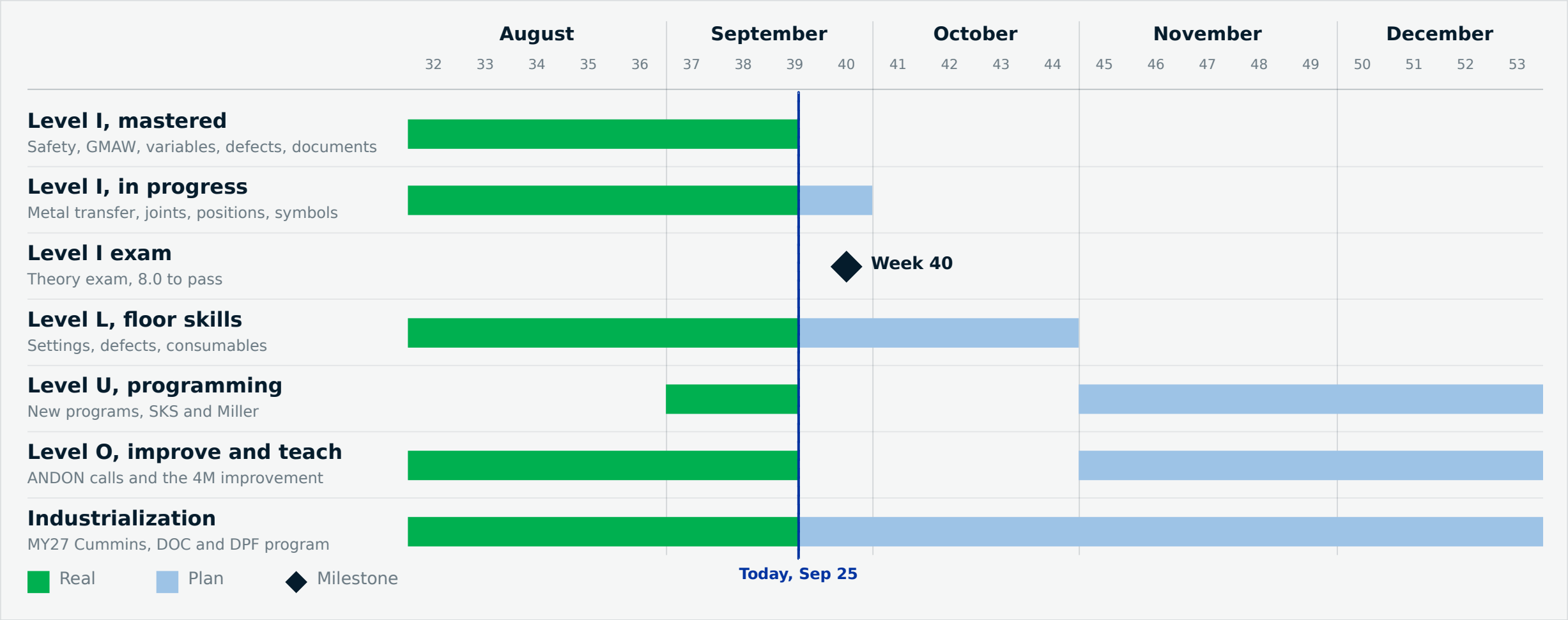
OBJECTIVE Become a specialist in robotic welding and advanced technical support.

LEVEL		WHAT IT COVERS	HOW IT IS VALIDATED	TIME	METHOD
I	Understands	Safety, GMAW equipment, metal transfer, process variables, discontinuities, joints, positions, symbols and control documents.	Theory exam, 8.0 to pass	4 h	Classroom session
L	With support	Basic robotic and dimensional adjustments, checking settings against the WPS, finding defects and changing consumables.	Validation on the floor	8 h	Robotic cell, on the floor (Gemba)
U	Alone	Creating new welding programs in SKS and setting their parameters (SKS and Miller).	Practical exam	8 h	Robotic cell, on the floor (Gemba)
O	Improves and teaches	Training level 2 and 3 staff, improvement ideas, problem solving (8D, Ishikawa, 5W+2H), GD&T, ANDON and Lean.	Implementation and training, validated	20 h	80-hour follow-up: TM, WAVE or PSIF



Level I exam next, in week 40

Gantt | Technical and practical skills, done and next



Level I: five topics mastered, four still open

What I learned

MASTERED

Safety	Working at the cell without risk, always with LOTOTO (lockout, tagout, tryout).
GMAW equipment	Every part of a GMAW welding system, and how they work together.
Process variables	Keeping the essential variables under control, because they decide weld quality.
Discontinuities	Reading the customer's weld requirements, spotting NG welds and finding their cause.
Control documents	WPS, PQR and parameter sheets: the approved recipe for every weld.

STILL OPEN

- > Metal transfer modes
- > Joint geometry
- > Welding positions
- > Welding symbols

I close them before the exam in week 40.

HOW I LEARNED IT 616 pages of manual and standard, a practice exam, and daily failures at the cells.



Eleven skills achieved, six in progress

Progress on the 23 skills of the program



ON THE FLOOR TODAY

- Changing welding consumables
- Adjusting the key welding settings within their ranges
- Closing ANDON calls

Next: Level I exam, week 40.

Most of my time goes to a program that is 90% ready

My time on the floor, and where the program stands

60%

of my time: industrialization, with Engineering



40%

Support at the robotic cells: stops, adjustments and ANDON calls.

THE PROJECT

Industrialization and successful PPAP runs of the MY27 Cummins DOC and DPF program, together with Engineering.

PROGRAM PROGRESS

90%

ready to launch

Start of production planned for January 2027.



Improvement idea: from a manual booth to the 4M cell

Improvement idea and implementation | Done

THE PROBLEM

Long weld seams were hard to control by hand, the welder worked in uncomfortable positions, and welding alone took 146 s per part.

THE IDEA

Move the process to 4M, an automatic robotic cell, with the same fixture.

IMPLEMENTED

Running in 4M today. The robot welds, and the team member only loads and unloads.



Same part, 47% less cycle time

Improvement | Results, in seconds per part

CYCLE TIME PER PART

Manual welding process



4M robotic cell



■ Load and unload ■ Welding by hand ■ Welding by robot

Same fixture, so loading and unloading take the same time. The gain is all in the welding.

-47%

cycle time, 174 s to 93 s

-56%

welding time, 146 s to 65 s

81 s

saved on every part



What I would change in how setters learn

Final project | Training process update, a proposal

01

Reviewed

616 pages of training manual and customer standard, page by page, against what Level I asks.

02

Found

Gaps: topics the exam asks about that the material barely covers, and an error in one key table.

03

Rebuilt

The schedule, in the plant's own format, so plan and reality can be compared week by week.

PROPOSAL

Use the corrected material and the new schedule with the next group of setters.

Next: from knowing it to doing it alone

What comes next, until January

NEXT LEVELS

- L** Validate settings and find defects at the cell, with support.
October, validation on the floor

- U** Create and set up robot programs on my own (SKS and Miller).
November to December

- O** Solve problems, share what I learn, keep improving.
November to December, already started with 4M

WHAT I NEED

- Level I exam confirmed, week 40

- Supervised time programming robots in SKS and Miller

- Your feedback on the training proposal



Conclusions and next steps

What is in place, and what comes next

IN PLACE

I studied all of Level I and mastered five of its nine topics, and I practiced the five Level L skills with my tutor and the technicians.

A manual weld moved to the 4M robotic cell: 47% less cycle time, less strain.

A reviewed study path, and a schedule that shows plan versus reality.

NEXT STEPS

Close the four open Level I topics and pass the exam in week 40.

Validate Level L on the floor in October, then reach U and O before January.

Keep measuring 4M, and find the next weld or process to automate.

I came to learn robotic welding, and I am already improving it.

The background of the slide is a dark blue image of a tunnel. Multiple bright light trails, likely from a train, streak across the frame from the top left towards the bottom right, creating a sense of motion and depth.

TENNECO

Thank you

Diego Adair de León Márquez | Engineering Department